F79SP Session 4: Simple Random Walks

1 Simple random walks

Definition 1.1. *Let $S_0 = a \in \mathbb{Z}$ and $\xi_1, \xi_2, \dots$ be i.i.d. r.v's with:*

$$\begin{aligned} \mathbf{P}(\xi_i = 1) &= p \qquad\qquad (0 < p < 1) \\ \mathbf{P}(\xi_i = -1) &= q = 1 - p \\ (\Rightarrow \mathbf{P}(\xi_i = k) &= 0 \quad \textit{for all other integers } k) \end{aligned}$$

Let $S_n = a + \sum_{i=1}^{n} \xi_i$ for $n > 0$. Then, $\{S_n\}_{n \geq 0}$ is called a ***simple random walk*** *on the* ***integer state space*** $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$.
Note: *A simple random walk is called* ***symmetric*** *if $p = q = \frac{1}{2}$.*

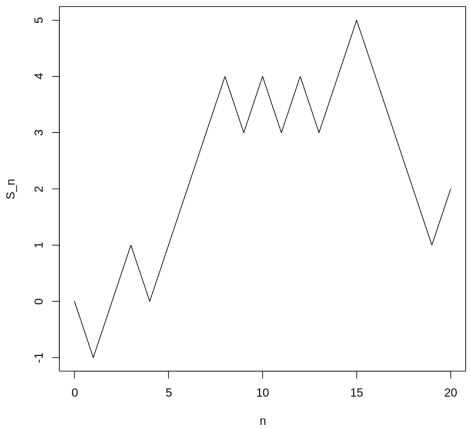


Figure 1: A realisation of symmetric simple random walk ($p = 0.5$).

Note that:

$$\begin{aligned} \mu &\equiv \mathbf{E}\,\xi_i = p - q \\ \sigma^2 &\equiv \mathbf{Var}\xi_i = \mathbf{E}\,\xi_i^2 - \mu^2 = 1 - (p-q)^2 = 4pq \leq 1, \end{aligned}$$

and note that $4pq = 1$ if and only if $p = q = 1/2$, and $4pq < 1$, otherwise. Recall that we then have

$$\begin{aligned} \mathbf{E}\,S_n &= a + n\mu, \\ \mathbf{Var}S_n &= n\sigma^2 \end{aligned}$$

as in the general case, where μ and σ^2 are now given as above.

Time and space homogeneity

A simple random walk (SRW) has below properties:

- Temporally/Time homogeneous: $\mathbf{P}(S_n = j|S_0 = a) = \mathbf{P}(S_{n+m} = j|S_m = a)$ for every $m \in \mathbb{Z}^+$.
- Spatial/Space homogeneous: $\mathbf{P}(S_n = j|S_0 = a) = \mathbf{P}(S_n = j+b|S_0 = a+b)$ for every $b \in \mathbb{Z}$ (with abuse of notation).

Postnote on time homogeneity:

- Says we are currently at time $n = 0$ with $S_0 = a$, then $S_n = a + \sum_{i=1}^{n} \xi_i$ (sum of n many i.i.d. ξ_i's).
- Says we are currently at time $n = m$ with $S_m = a$, then $S_{n+m} = S_m + \sum_{i=m+1}^{n+m} \xi_i = a + \sum_{i=m+1}^{n+m} \xi_i$ (sum of n many i.i.d. ξ_i's)

Exact formula for simple random walks with $a = 0$

Motivational question: Determine $\mathbf{P}(S_{10} = 0)$ and $\mathbf{P}(S_{10} = 1)$.

General observation: Let $n \geq 0$. S_{2n} is always even and S_{2n+1} is always odd. Hence, $\mathbf{P}(S_{2n} = 2m + 1) = 0$ and $\mathbf{P}(S_{2n+1} = 2m) = 0$.

- For $-n \leq m \leq n$, $\mathbf{P}(S_{2n} = 2m) = \binom{2n}{n+m} p^{n+m} q^{n-m}$.
- For $-n - 1 \leq m \leq n$, $\mathbf{P}(S_{2n+1} = 2m + 1) = \binom{2n+1}{n+m+1} p^{n+m+1} q^{n-m}$.

Proof (for $\mathbf{P}(S_{2n} = 2m)$).

- Let X denote the number of $(+1)$'s in the first $2n$ steps and Y the number of (-1)'s.

Asymptotics for the probability to return to zero for random walks

Consider the **symmetric simple random walk**, defined by $p = q = 1/2$ and starting at $S_0 = 0$. We are interested with the intensity of the SRW revisiting 0 in long run.

1. Using the exact formula derived earlier with $m = 0$, we have for every $n \in \mathbb{Z}^+$:

$$\mathbf{P}(S_{2n} = 0) = \binom{2n}{n} \frac{1}{2^{2n}} = \frac{(2n)!}{(n!)^2} \frac{1}{2^{2n}}.$$

   Now use **Stirling's approximation**: $n! \sim e^{-n} n^n \sqrt{2\pi n}$ as $n \to \infty$ to obtain:

$$\begin{aligned} \mathbf{P}(S_{2n} = 0) &\sim \frac{e^{-2n}(2n)^{2n}\sqrt{4\pi n}}{e^{-2n} n^{2n} (2\pi n)} \frac{1}{2^{2n}} \\ &= \frac{1}{\sqrt{\pi n}}, \end{aligned}$$

   valid for all large n.

   Note that this probability declines in proportion to $1/\sqrt{n}$, exactly as we would expect (see below).

2. For n moderate to large (and odd or even), find a reasonable approximation to the entire distribution of S_n. We have

$$\mathbf{E}\, S_n = 0, \qquad \mathbf{Var} S_n = n,$$

   and so the distribution of S_n is approximately normal with this mean and variance (i.e. with standard deviation $\sqrt{n}$).

3. Assume now that $p \neq q$ (this means that either $0 < p < 1/2$ or $1/2 < p < 1$). Then

$$\mathbf{P}(S_{2n} = 0) = \binom{2n}{n} p^n q^n = \binom{2n}{n} \frac{1}{2^{2n}} \cdot (4pq)^n = \frac{(2n)!}{(n!)^2} \frac{1}{2^{2n}} \cdot (4pq)^n.$$

   Use again **Stirling's approximation**, to deduce

$$\mathbf{P}(S_{2n} = 0) \sim \frac{e^{-2n}(2n)^{2n}\sqrt{4\pi n}}{e^{-2n} n^{2n} (2\pi n)} \frac{1}{2^{2n}} \cdot (4pq)^n = \frac{1}{\sqrt{\pi n}} \cdot (4pq)^n, \tag{1}$$

   so now the probabilities decay exponentially fast!